

Source-Domain Confounding as the Primary Driver of Overestimated Ames Mutagenicity QSAR Performance: A Multi-Level Evaluation Across Three Genotoxicity Endpoints

Seungmin Yoon, Sanghun Kim, and Hyunpyo Jeon*

Department of Biosafety, Kyungsoong University, Busan, Republic of Korea

E-mail: hpjeon@ks.ac.kr

Abstract

Quantitative structure–activity relationship (QSAR) models for Ames mutagenicity are increasingly used in ICH M7 assessments,[?] yet reported performance varies greatly depending on evaluation methodology. Here we present a systematic evaluation of 375 model configurations across five genotoxicity datasets—Ames mutagenicity ($n = 13,560$), in vitro chromosome aberration ($n = 1,619$), in vivo micronucleus ($n = 2,153$), and two negative-sampled variants—using seven algorithms, five feature representations, and three preprocessing scenarios. Using a single locked configuration (XGBoost, compact descriptors, unprocessed SMILES) evaluated at three levels of increasing stringency, we show that for Ames, performance decreased from random-split cross-validation ($\text{MCC} = 0.670$) to scaffold-split cross-validation (0.624 , $\Delta = 0.046$) to cross-source-domain leave-domain-out in the drug-to-industrial direction (0.234). Source-domain confounding—arising from the mixing of drug-like compounds (positive rate 58%) and industrial chemicals (positive rate 3%)—drove a scaffold-to-LDO

MCC drop of $\Delta = 0.390$, substantially exceeding the split-method effect. A naive source-domain classifier achieved $\text{MCC} = 0.608$ without any structural information. However, algorithm robustness under domain shift was not uniform: Random Forest maintained $\text{MCC} = 0.325$ in the most challenging cross-domain direction where XGBoost collapsed to 0.122. In vivo micronucleus prediction showed some ranking ability ($\text{ROC-AUC} = 0.860$, $\text{PR-AUC} = 0.237$) after salt stripping but lacked reliable classification utility ($\text{MCC } 95\% \text{ CI crossing zero}$). We propose multi-level evaluation with source-domain awareness as a reporting standard for Ames mutagenicity QSAR and provide an open-source pipeline for reproducibility.

Acknowledgement

dd

Introduction

Regulatory Context

The ICH M7(R2) guideline recommends the use of two complementary in silico approaches for assessing the mutagenic potential of pharmaceutical impurities: a rule-based expert system and a statistical QSAR model.[?] This regulatory framework has driven considerable interest in developing predictive models for the Ames bacterial reverse mutation assay,^{??} with extensions to other genotoxicity endpoints including in vitro chromosome aberration (CA) and in vivo micronucleus (MN) tests.^{??} Reliable QSAR models for these endpoints could reduce animal testing, accelerate safety assessments, and support the 3Rs principle.[?]

The Overestimation Problem

Published Ames QSAR models report a wide range of performance metrics, with accuracies typically spanning 0.80–0.90 and MCC values of 0.60–0.80.^{1,2,3,4} However, these numbers arise from different data splits, datasets, preprocessing pipelines, and evaluation protocols, making direct comparison difficult.

Several factors can inflate apparent model performance. First, random data splitting allows structurally similar compounds (sharing the same Murcko scaffold⁵) to appear in both training and test sets, creating an optimistic estimate of generalization to novel structures. Scaffold-based splitting has been recommended as a more realistic evaluation strategy,^{6,7} but the magnitude of performance difference between random and scaffold splits has not been systematically quantified across multiple genotoxicity endpoints.

Second, most large Ames datasets aggregate compounds from distinct sources—typically pharmaceutical compound libraries and industrial chemical registries.^{8,9} These sources differ in composition and mutagenicity prevalence. When these source domains are mixed in training and test sets, a model may partly learn source-domain membership rather than toxicity mechanisms—a form of confounding that has been noted^{10,11} but, to our knowledge, never systematically measured.

Third, preprocessing choices such as salt stripping, metal compound removal, and charge normalization can substantially alter molecular representations,^{12,13} yet these steps are often undocumented or applied uniformly without assessing endpoint-specific effects.

Gaps in the Current Literature

Most published studies evaluate a single endpoint using a single split method and focus their analysis on algorithm comparison.^{14,15} Cross-domain generalization is rarely tested. Statistical significance of model differences is seldom reported.¹⁶ The relative contributions of split method, source-domain composition, and preprocessing choices to apparent model performance have not been disentangled within a single consistent experimental framework.

Contributions

This work makes the following contributions:

1. We introduce a three-level evaluation framework that progressively increases evaluation rigor—in-domain scaffold split, random-versus-scaffold comparison, and cross-source-domain leave-domain-out—applied to a single locked model configuration for fair comparison.
2. We demonstrate that, for the Ames mutagenicity benchmark studied here, source-domain confounding arising from the mixing of drug-like and industrial compounds with a 19.5-fold prevalence difference drove a scaffold-to-LDO MCC drop of 0.390, substantially exceeding the split-method effect ($\Delta = 0.046$).
3. We show that while in-domain algorithm differences were small across all seven algorithms (MCC range 0.063 under a fixed feature/preprocessing configuration), cross-domain robustness varied meaningfully—Random Forest maintained $\text{MCC} = 0.325$ under domain shift where XGBoost collapsed to 0.122.
4. We demonstrate that in vivo micronucleus models showed some ranking signal ($\text{ROC-AUC} = 0.860$, $\text{PR-AUC} = 0.237$) after salt stripping but lacked reliable classification utility, with MCC confidence intervals crossing zero.
5. We demonstrate endpoint-specific preprocessing effects, with salt stripping improving in vivo best-model MCC by +0.22 while having mixed or negative effects on other endpoints.
6. We provide an open-source pipeline with all experimental configurations, split seeds, and locked results tables for full reproducibility.

Materials and Methods

Datasets

Five genotoxicity datasets were evaluated (Table ??). The three primary datasets comprise Ames bacterial reverse mutation ($n = 13,560$ after conflict resolution), in vitro chromosome aberration ($n = 1,619$), and in vivo micronucleus ($n = 2,153$). Two additional negative-sampled variants (in vitro sampling, $n = 345$; in vivo sampling, $n = 231$) were evaluated as exploratory supplementary analyses with reduced negative compounds.

The Ames dataset was compiled from pharmaceutical (drug-like) and industrial chemical sources. Compounds were annotated with source-domain labels based on their originating data source. We use the term “source domain” throughout to emphasize that this annotation reflects data provenance rather than a computed chemical classification, though the two sources differ substantially in mutagenicity prevalence—a pattern consistent with distinct source-domain composition. Drug-sourced compounds ($n = 6,416$) had a positive rate of 58.3%, while industrial-sourced compounds ($n = 7,144$) had a positive rate of 3.0%, a 19.5-fold difference. In vitro and in vivo datasets lacked source-domain annotations.

Label conflicts (compounds with contradictory labels across sources) were resolved conservatively by exclusion: 169 for Ames, 3 for in vitro, and 2 for in vivo.

Cross-endpoint overlap analysis revealed substantial compound sharing: 97.5% of in vitro compounds and 92.1% of in vivo compounds appeared in the Ames dataset, with moderate label concordance (Cohen’s $\kappa = 0.14$ – 0.20). This overlap indicates that the three endpoints largely probe the same compound universe with different biological readouts, a point we return to in the Discussion.

All datasets were split into training (80%) and test (20%) partitions using scaffold-based splitting[?] with zero scaffold group overlap. Murcko scaffolds[?] were generated using RD-Kit,[?] with acyclic compounds clustered by agglomerative clustering on Morgan fingerprint[?] similarity.

Preprocessing Scenarios

Three preprocessing scenarios were applied independently to each dataset:

Raw (raw_all): Original SMILES used without modification.

Salt-stripped (salt_stripped): The largest molecular fragment was retained using RDKit’s `LargestFragmentChooser`,[?] followed by charge normalization. This modified 45% of Ames, 81% of in vitro, and 83% of in vivo SMILES.

Metal-removed (no_metal): Compounds containing metal atoms were excluded.

Feature Representations

Five feature representations were evaluated:

Compact (138 features): 13 physicochemical descriptors (MW, LogP, TPSA, HBD, HBA, rotatable bonds, fraction C_{sp3}, aromatic ring count, ring count, heteroatom count, heavy atom count, formal charge, valence electrons) and 125 structural alert features based on Benigni–Bossa SA1–SA31,[?] Kazius toxicophores,[?] and ISS Ames alerts.

Broad FP-512/1024/2048 (266/394/522 features): Compact features plus 128/256/384 Morgan fingerprint bits[?] selected by mutual information[?] on the training set only. Bits with prevalence below 1% or above 99% were excluded before MI ranking.

Graph: Molecular graphs with 9-dimensional atom features and normalized adjacency matrices, used exclusively with the GCN model.

Algorithms

Seven classification algorithms were evaluated: XGBoost[?] and LightGBM[?] (gradient boosting), Random Forest[?] (bagging ensemble), SVM with RBF kernel[?] and Logistic Regression[?] (both with standard scaling), MLP[?] (128–64 hidden units, early stopping), and a three-layer Graph Convolutional Network[?] with learnable weights, LayerNorm, and global mean pooling. Class imbalance was addressed using `scale_pos_weight` or `class_weight='balanced'`.

Hyperparameter tuning was performed using `RandomizedSearchCV` with `GroupKFold` (scaffold groups).[?]

Three-Level Evaluation Framework

To ensure fair comparison across evaluation levels, we defined a locked reference configuration: XGBoost with compact features and raw SMILES. All three levels were evaluated using this single configuration:

Level 1—In-domain scaffold split: Trained on the scaffold-disjoint training partition, evaluated on the held-out test partition. MCC[?] with 500-iteration bootstrap 95% CIs, plus ROC-AUC, PR-AUC, balanced accuracy, sensitivity, specificity, and Brier score.[?]

Level 2—Random versus scaffold split comparison: Both 5-fold stratified random CV (5 repeats) and 5-fold scaffold-grouped CV (5 repeats) on the full dataset.

Level 3—Cross-source-domain leave-domain-out (LDO): For Ames, trained on one source domain and tested on the other, using the locked configuration plus two additional algorithms (LightGBM, Random Forest) to assess algorithm-specific robustness under domain shift.

Additional Analyses

Full grid evaluation: Beyond the locked configuration, all 375 combinations were evaluated on the scaffold-split test set. The 375 total comprises $5 \text{ datasets} \times 3 \text{ preprocessing scenarios} \times (6 \text{ tabular algorithms} \times 4 \text{ feature modes} + 1 \text{ GNN}) = 5 \times 3 \times 25 = 375$.

Applicability domain (AD): Tanimoto similarity on binary 1024-bit Morgan fingerprints.[?] Threshold ≥ 0.3 defined AD membership.

Feature importance: SHAP values[?] (`TreeExplainer`) for tree-based models on the three primary endpoints.

Learning curves: Training set fractions 10%–100% with 5-fold scaffold CV, using the locked configuration. Learning curves for alternative configurations (e.g., salt-stripped broad

FP) were not computed and represent a limitation.

Statistical comparison: McNemar’s test[?] with continuity correction for pairwise comparison of the top two models per endpoint.

Threshold tuning: Out-of-fold probability-based threshold optimization using 5-fold GroupKFold.

Results

Best Observed Performance Across the Full Grid

Table ?? presents the best observed performance for each primary endpoint across all 375 configurations evaluated on the scaffold-split test set. These values represent an upper bound of achievable performance under extensive model/feature/preprocessing search and should be interpreted as a descriptive benchmark, not as unbiased generalization estimates. The confirmatory analysis uses the locked configuration (Table ??).

For Ames, XGBoost with broad FP-2048 achieved $\text{MCC} = 0.555$ (95% CI: 0.522–0.588), $\text{ROC-AUC} = 0.850$, and $\text{PR-AUC} = 0.765$. For in vitro CA, LightGBM with salt-stripped broad FP-1024 achieved $\text{MCC} = 0.297$ (95% CI: 0.131–0.450) and $\text{PR-AUC} = 0.278$. For in vivo MN, XGBoost with salt-stripped broad FP-512 achieved $\text{MCC} = 0.311$ (95% CI: -0.014 to 0.571), $\text{ROC-AUC} = 0.860$, and $\text{PR-AUC} = 0.237$ (baseline $\text{PR} = 0.028$).

Under a fixed feature and preprocessing configuration (compact features, raw SMILES), all seven algorithms were compared for Ames on the scaffold-split test set (Table ??). MCC ranged from 0.430 (Logistic Regression) to 0.492 (Random Forest), a span of 0.063 across all algorithms. The five tabular models excluding Logistic Regression spanned only 0.023 (0.470–0.492). GCN achieved 0.445. Pairwise McNemar tests[?] between the top two models yielded no significant differences (Ames $p = 0.39$; in vivo $p = 0.72$). However, this in-domain equivalence did not extend to cross-domain settings (Section ??).

Evaluation Rigor Cascade (Locked Configuration)

Table ?? and Figure ?? present results exclusively from the locked configuration (XGBoost, compact features, raw SMILES), ensuring that observed differences reflect only the evaluation method.

For Ames, random-split CV yielded $\text{MCC} = 0.670 \pm 0.010$, consistent with published benchmarks using similar strategies.?? Scaffold-split CV yielded $\text{MCC} = 0.624 \pm 0.049$, a reduction of 0.046. The scaffold-split held-out test yielded $\text{MCC} = 0.470$ (95% CI: 0.434–0.502).

Cross-source-domain LDO revealed a substantially larger performance collapse. Drug $\rightarrow$ industrial yielded $\text{MCC} = 0.234$; industrial $\rightarrow$ drug yielded $\text{MCC} = 0.122$.

Comparing magnitudes using absolute MCC differences: the source-domain gap (scaffold CV 0.624 $\rightarrow$ LDO drug $\rightarrow$ industrial 0.234, $\Delta = 0.390$) substantially exceeded the split-method gap (random CV 0.670 $\rightarrow$ scaffold CV 0.624, $\Delta = 0.046$). We note that the LDO comparison involves additional confounds beyond domain membership (directional asymmetry, different test set sizes and class distributions), so $\Delta = 0.390$ should be interpreted as reflecting the combined effect of domain shift plus these structural differences, rather than a pure measurement of domain confounding alone. Nevertheless, the magnitude difference between the two effects is unambiguous in this benchmark.

For in vitro CA, random and scaffold splits produced nearly identical results (MCC 0.180 versus 0.189, $\Delta = -0.009$). For in vivo MN, random split produced $\text{MCC} = 0.181$ versus scaffold split $\text{MCC} = 0.104$ ($\Delta = 0.077$), though both values represent near-random prediction given their standard deviations (0.094 and 0.111).

Source-Domain Confounding and Algorithm-Specific Robustness

The Ames dataset exhibited severe source-domain confounding (Figure ??). Drug-sourced compounds ($n = 6,416$, positive rate 58.3%) and industrial-sourced compounds ($n = 7,144$, positive rate 3.0%) differed by a factor of 19.5 in mutagenicity prevalence. A naive classifier

predicting positive for drug-sourced and negative for industrial-sourced compounds achieved $\text{MCC} = 0.608$ and balanced accuracy = 0.834, comparable to the random-split CV MCC of 0.670, without any structural information.

LDO with three algorithms revealed two findings (Table ??). First, drug $\rightarrow$ industrial generalization was consistent across algorithms (MCC 0.232–0.235), suggesting sufficient positive examples in the drug training set for all models.

Second, industrial $\rightarrow$ drug generalization exposed substantial algorithm-specific differences. XGBoost achieved $\text{MCC} = 0.122$ (sensitivity 0.062), LightGBM 0.134 (sensitivity 0.081), while Random Forest maintained $\text{MCC} = 0.325$ (sensitivity 0.466). This contrast is notable because these algorithms performed comparably in the in-domain evaluation (Section 3.1). We hypothesize that Random Forest’s superior cross-domain robustness arises from feature bagging,[?] which reduces reliance on domain-correlated feature subsets. This finding nuances the in-domain algorithm equivalence: algorithm selection may matter more in deployment scenarios where the target chemical space differs from the training domain.

Preprocessing Impact

Salt stripping effects were endpoint-specific (Figure ??). For in vivo MN, the best salt-stripped model achieved $\text{MCC} = 0.311$, substantially outperforming the best raw model ($\text{MCC} = 0.086$), an improvement of +0.225. Salt fragments were present in 83% of in vivo SMILES. Among salt-containing in vivo compounds, the positive rate was 2.8%, while non-salt compounds showed 7.1%, suggesting that salt fragments were disproportionately present in negative-labeled compounds and may have interfered with learning toxicophore patterns.

For Ames, salt stripping produced mixed effects ($\Delta = -0.03$ to +0.02). For in vitro CA, salt stripping was generally detrimental (mean MCC decreased from 0.081 to 0.056). This endpoint specificity challenges the common practice of applying uniform preprocessing[?] and argues for per-endpoint optimization.

In Vivo Micronucleus: Ranking Signal Versus Classification Utility

The in vivo MN results require careful interpretation. The best model (salt-stripped XGBoost, broad FP-512) achieved ROC-AUC = 0.860 and PR-AUC = 0.237 (versus a baseline PR of 0.028, representing an 8.5-fold enrichment over random). This suggests some ranking ability. However, MCC = 0.311 with 95% CI crossing zero (-0.014 to 0.571) indicates unreliable classification at any fixed threshold, reflecting the extreme class imbalance (training positive rate 3.7%; test set: 12 positives out of 431, 2.8%).

The learning curve for the locked configuration (compact features, raw SMILES) showed flat MCC (≈ 0.055) from 10% to 100% of training data (Figure ??). We note that this curve was not computed for the best-performing configuration (salt-stripped, broad FP-512), so the flat profile may partly reflect the baseline feature and preprocessing choices. The salt stripping effect ($+0.225$ for the best model) demonstrates that preprocessing artifacts can mask whatever structural signal exists in the data.

We frame the in vivo finding as follows: current models show ranking signal for in vivo micronucleus genotoxicity after appropriate preprocessing, but threshold-dependent classification utility is unreliable given extreme class imbalance and small positive test counts. In vivo QSAR may have value as a component of tiered screening[?] but should not serve as a standalone binary classifier with current 2D descriptors.

Feature Importance

SHAP analysis[?] revealed consistent patterns across endpoints (Figure ??). Physicochemical descriptors dominated the top-5 features for all three primary endpoints: the aggregate genotoxicity alert count (mean $|\text{SHAP}| = 0.82$) and MW (0.57) for Ames; LogP (0.58) and fraction $C_{\text{sp}3}$ (0.40) for in vitro; MW (1.00) and valence electrons (0.62) for in vivo. Individual fingerprint bits were absent from the top-5 for all endpoints despite comprising 50–75% of features in the broad modes. Comparing mean MCC across all models and preprocessing scenarios for the Ames endpoint, compact features yielded mean MCC = 0.448, while broad

FP-2048 yielded 0.523 ($\Delta = +0.075$), indicating that fingerprints provided distributed but individually weak supplementary information.

Learning Curves

Learning curves for the locked configuration (Figure ??) revealed three patterns: Ames MCC saturated early (0.543 at 10% of data, 0.640 at 100%); in vitro showed a noisy upward trend ($0.070 \rightarrow 0.170$); in vivo was flat (≈ 0.055 throughout). As noted in Section ??, the in vivo curve reflects the locked configuration only.

Discussion

The Overestimation Hierarchy in This Benchmark

Using a locked model configuration, we observed the following hierarchy of effects on apparent Ames QSAR performance:

Source-domain composition $\gg$ Data splitting method $\approx$ In-domain algorithm choice

The source-domain gap (scaffold CV $\rightarrow$ LDO, $\Delta = 0.390$) substantially exceeded the split-method gap (random $\rightarrow$ scaffold CV, $\Delta = 0.046$), which was comparable to the in-domain algorithm spread (MCC range = 0.063 across all seven algorithms under fixed features and preprocessing).

Three caveats apply. First, this hierarchy was established on a single Ames benchmark where source-domain labels were available; other datasets may exhibit different confounding structures. Second, the LDO $\Delta = 0.390$ reflects the combined effect of domain shift plus structural confounds (asymmetric test sizes, different class distributions per domain) and should be interpreted as an upper bound. Third, algorithm robustness was not equivalent under domain shift (Section ??), so algorithm choice may matter substantially more in cross-domain deployment than the in-domain comparison suggests.

Source-Domain Confounding: Provenance Versus Chemistry

A key question is whether the observed confounding reflects genuine chemical domain differences or data-source provenance artifacts. Our source-domain labels are based on originating data sources, not computed chemical descriptors. However, the 19.5-fold prevalence difference and the naive classifier’s MCC of 0.608 are consistent with distinct source-domain composition being highly correlated with mutagenicity outcomes. Prior work has noted that pharmaceutical and industrial chemical libraries occupy different regions of chemical property space,^{27, 28} which our prevalence data corroborates indirectly. A more rigorous analysis would include chemical space visualization (e.g., UMAP on molecular descriptors²⁹), matched-subset comparisons controlling for chemical similarity across sources, and source-adjusted performance estimates. These analyses would strengthen the causal interpretation and represent a priority for future work.

In Vivo Prediction: Implications for Tiered Testing

The dissociation between ranking ability ($\text{ROC-AUC} = 0.860$, $\text{PR-AUC} = 0.237$) and classification reliability (MCC CI crossing zero) has practical implications. For ICH M7-style binary hazard classification,²⁷ in vivo MN QSAR does not achieve sufficient reliability with current 2D descriptors. However, the PR-AUC of 0.237 represents an 8.5-fold enrichment over baseline prevalence (2.8%), suggesting potential value in tiered screening strategies.²⁷

The strong preprocessing sensitivity (+0.22 MCC from salt stripping) indicates that reported in vivo QSAR performance depends critically on preprocessing choices that are often undocumented.²⁷ The biological plausibility of in vivo prediction limitations is high: micronucleus induction depends on ADME properties, metabolic activation, tissue distribution, and DNA repair capacity,^{27, 28} none of which are captured by 2D descriptors.

Cross-Endpoint Overlap

The high compound overlap between endpoints ($>90\%$ shared with Ames) indicates that performance differences across endpoints (Ames $MCC = 0.555$ versus in vivo $MCC = 0.311$) are unlikely to be primarily explained by different chemical spaces alone, but rather reflect inherent differences in endpoint predictability from 2D structure.[?] Multi-task learning approaches jointly predicting all endpoints may be a promising direction for future work.[?]

Preprocessing Deserves Systematic Treatment

Salt stripping effects were endpoint-specific: beneficial for in vivo ($+0.22$ best-model improvement), mixed for Ames (± 0.03), and generally detrimental for in vitro (-0.03 to -0.13). We hypothesize that salt stripping harms in vitro CA performance because some counterions are themselves relevant to the clastogenicity assay. A stratified analysis by salt-containing versus non-salt and metal versus non-metal compounds would directly test this hypothesis and represents a tractable extension. The QSAR data curation literature^{??} has emphasized the importance of standardization, but our results suggest that the optimal standardization strategy is endpoint-dependent.

Practical Recommendations

Based on our findings, we propose the following reporting standards for genotoxicity QSAR:

1. Use scaffold-based splitting for primary performance reporting.
2. When source-domain labels are available, report leave-domain-out performance and disclose source composition with per-source positive rates.
3. Optimize preprocessing per endpoint rather than applying uniform pipelines.
4. Report bootstrap CIs and applicability domain coverage.

5. For endpoints with extreme class imbalance, separately report threshold-free metrics (ROC-AUC, PR-AUC) and threshold-dependent metrics (MCC, balanced accuracy).
6. When claiming algorithm superiority, test under domain shift as well as in-domain conditions.

Limitations

(a) Source-domain confounding analysis was limited to Ames; other endpoints lacked source annotations. (b) Source-domain labels reflect provenance rather than computed chemical classification. (c) The in vivo learning curve used only the locked configuration, not the best-performing configuration. (d) A single scaffold split was used rather than repeated holdout. (e) The GCN did not include edge features or pre-training.[?] (f) No 3D descriptors, QM descriptors, or metabolite predictions were included. (g) Within-AD versus outside-AD performance stratification for LDO was not conducted. (h) McNemar tests were limited to pairwise top-two comparisons; comprehensive multiple-comparison correction[?] was not applied. (i) Table ?? reports best observed performance across 375 configurations evaluated on the test set, which introduces model-selection bias; the locked-configuration results (Table ??) provide unbiased estimates.

Conclusion

We present a multi-level evaluation of genotoxicity QSAR revealing that, for the Ames mutagenicity benchmark studied here, source-domain confounding arising from the mixing of drug-like and industrial chemicals drove a substantially larger performance gap ($\Delta = 0.390$ from scaffold CV to LDO) than the choice between random and scaffold splitting ($\Delta = 0.046$). Algorithm differences were small in the in-domain setting (MCC range = 0.063) but emerged under domain shift, with Random Forest showing better cross-domain robustness than gradient boosting. In vivo micronucleus models showed ranking signal but lacked reliable clas-

sification utility with current 2D descriptors. We recommend multi-level evaluation with source-domain awareness as a minimum reporting standard for Ames mutagenicity QSAR.

Data and Code Availability

The complete pipeline code (v12.1), configuration files, split seed assignments, and all locked results tables are available at [https://github.com/\[username\]/genotox-qsar-evaluation](https://github.com/[username]/genotox-qsar-evaluation) (DOI: [to be assigned upon acceptance]). The repository includes instructions for exact reproduction of all reported numbers, including the scaffold-split assignments (global seed = 42), preprocessing flags, and per-experiment metrics files.

Author Information

Corresponding Author

*E-mail: author@university.edu

Notes

The authors declare no competing financial interest.

Acknowledgments

[To be added]

References

.

Table 1: Dataset summary. Positive rate refers to the training partition. The 375 total configurations comprise 5 datasets $\times$ 3 preprocessing scenarios $\times$ (6 tabular models $\times$ 4 feature modes + 1 GNN).

Dataset	Assay	n	Train	Test	Pos. rate (%)	Scaffolds
Ames	Bacterial reverse mutation	13,560	10,848	2,712	29.1	1,691
In vitro CA	Chromosome aberration	1,619	1,295	324	10.4	353
In vivo MN	Micronucleus	2,153	1,722	431	3.7 / 2.8 ^a	410
In vitro CA (sampled) ^b	Neg. sampled CA	345	276	69	49.6	169
In vivo MN (sampled) ^b	Neg. sampled MN	231	184	47	33.3	129

^aTraining / test positive rate. ^bSupplementary exploratory datasets.

Table 2: Best observed performance per primary endpoint (scaffold-split test set, full grid). These represent best performance among all 375 evaluated configurations and should be interpreted as a descriptive benchmark, not unbiased generalization estimates. Bootstrap CIs reflect test-set sampling variability but do not account for model-selection variance. Confirmatory estimates from the locked configuration appear in Table ??.

Endpoint	Algo.	Preproc.	Features (n)	MCC [95% CI]	AUC	PR-AUC	Sens.	Spec.
Ames	XGB	raw	FP-2048 (522)	0.555 [.522,.588]	0.850	0.765	0.756	0.822
In vitro	LGBM	salt_str.	FP-1024 (394)	0.297 [.131,.450]	0.665	0.278	0.349	0.929
In vivo	XGB	salt_str.	FP-512 (266)	0.311 [−.014,.571]	0.860	0.237	0.319	0.983

PR-AUC baselines (prevalence): Ames 0.302, in vitro 0.124, in vivo 0.028.

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website.

- Table S1: Sampling endpoint results (exploratory; in vitro sampling $n = 69$ test, in vivo sampling $n = 47$ test)
- Table S2: Random versus scaffold split comparison for sampling endpoints
- Table S3: Full 375-experiment results with all metrics (CSV)
- Table S4: Hyperparameter tuning comparison (fixed vs. tuned parameters)
- Table S5: Applicability domain coverage by endpoint and configuration
- Table S6: Cross-endpoint compound overlap with label concordance

Table 3: Evaluation rigor cascade—locked configuration (XGBoost, compact features, raw SMILES). Levels are ordered by decreasing stringency of the in-domain assumption: Level 2 uses cross-validation on the full dataset (repeated 5×5), Level 1 uses a single held-out test set, and Level 3 trains and tests across source domains. All values are from the same locked configuration.

Level	Evaluation method	Ames MCC	In vitro MCC	In vivo MCC
2a	Random-split 5×5 CV	0.670 ± 0.010	0.180 ± 0.053	0.181 ± 0.094
2b	Scaffold-split 5×5 CV	0.624 ± 0.049	0.189 ± 0.084	0.104 ± 0.111
1	Scaffold-split test	0.470 [.434,.502]	0.114 [−.023,.270]	0.071 [−.029,.285]
3a	LDO: drug → industrial	0.234	—	—
3b	LDO: industrial → drug	0.122	—	—

Level 2: mean ± SD (25 folds). Level 1: MCC [95% bootstrap CI]. Level 3: Ames only (other endpoints lack source-domain labels). Levels ordered 2→1→3 to show progression from within-distribution CV through single-split held-out to cross-domain transfer.

Table 4: Leave-domain-out results by algorithm (Ames).

Algorithm	Drug → Ind. MCC	Ind. → Drug MCC	Ind. → Drug Sens.
XGBoost	0.234	0.122	0.062
LightGBM	0.232	0.134	0.081
Random Forest	0.235	0.325	0.466

Drug: $n = 6,416$, pos. rate = 58.3%. Industrial: $n = 7,144$, pos. rate = 3.0%. Prevalence ratio = 19.5-fold.
Naive source classifier: MCC = 0.608, balanced accuracy = 0.834.

- Table S7: SHAP top-20 features per endpoint

All supplementary CSV files are also available in the code repository.

Table 5: All-algorithm comparison under locked features (compact, raw SMILES, Ames scaffold-split test set).

Algorithm	MCC	ROC-AUC
Random Forest	0.492	0.838
ANN (MLP)	0.489	0.834
LightGBM	0.479	0.833
SVM (RBF)	0.470	0.832
XGBoost	0.470	0.825
GCN	0.445	0.815
Logistic Regression	0.430	0.800

Range across all 7 algorithms: 0.063. Range excluding LR and GCN: 0.023.

TOC Graphic

Some journals require a graphical entry for the Table of Contents. This should be laid out “print ready” so that the sizing of the text is correct. Inside the tocentry environment, the font used is Helvetica 8 pt, as required by *Journal of the American Chemical Society*.

The surrounding frame is 9 cm by 3.5 cm, which is the maximum permitted for *Journal of the American Chemical Society* graphical table of content entries. The box will not resize if the content is too big: instead it will overflow the edge of the box.

This box and the associated title will always be printed on a separate page at the end of the document.

Figure 1. Three-level evaluation framework

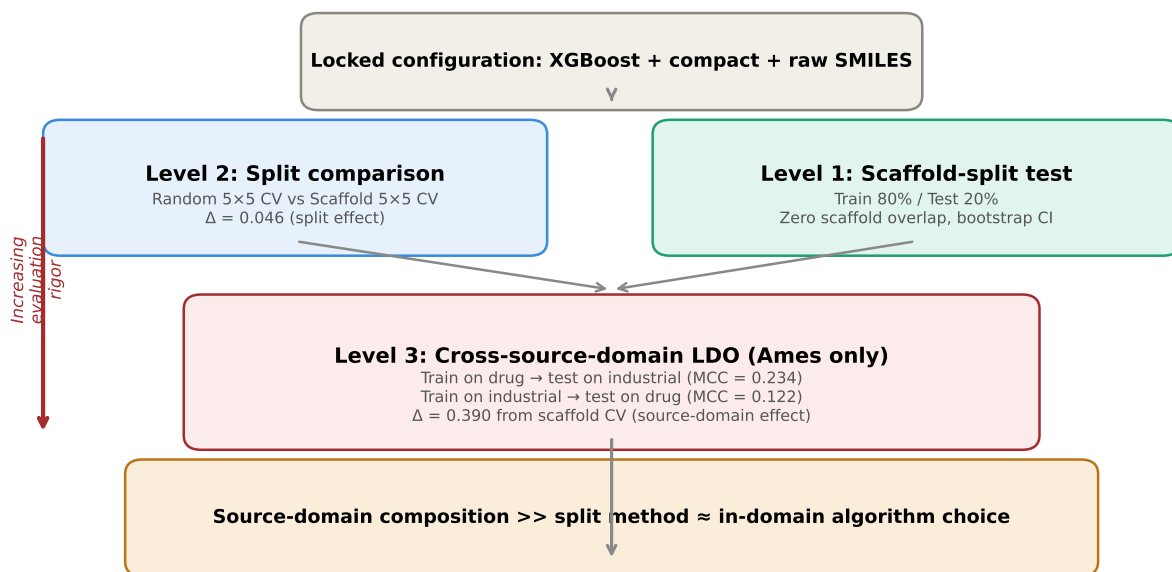


Figure 1: Three-level evaluation framework for genotoxicity QSAR. All three levels use the same locked configuration (XGBoost, compact features, raw SMILES) to enable fair comparison of MCC across evaluation methods. The overestimation hierarchy summarizes the relative magnitudes observed in this benchmark.

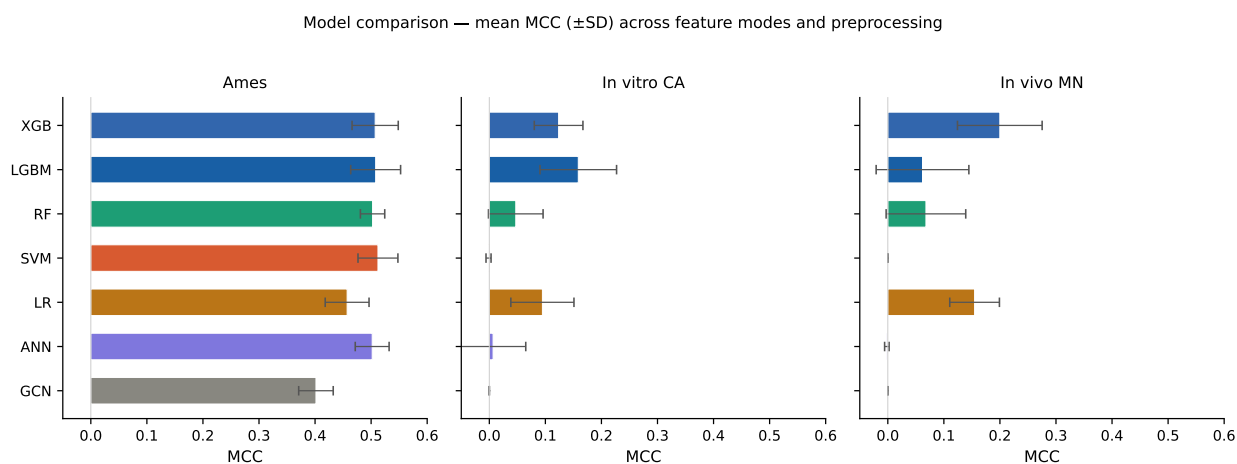


Figure 2: Mean MCC ($\pm$ SD) by algorithm across all feature modes and preprocessing scenarios, grouped by endpoint. Error bars represent ± 1 SD. Within each endpoint, the top five tabular models show minimal performance spread (Ames range = 0.023 excluding LR and GCN).

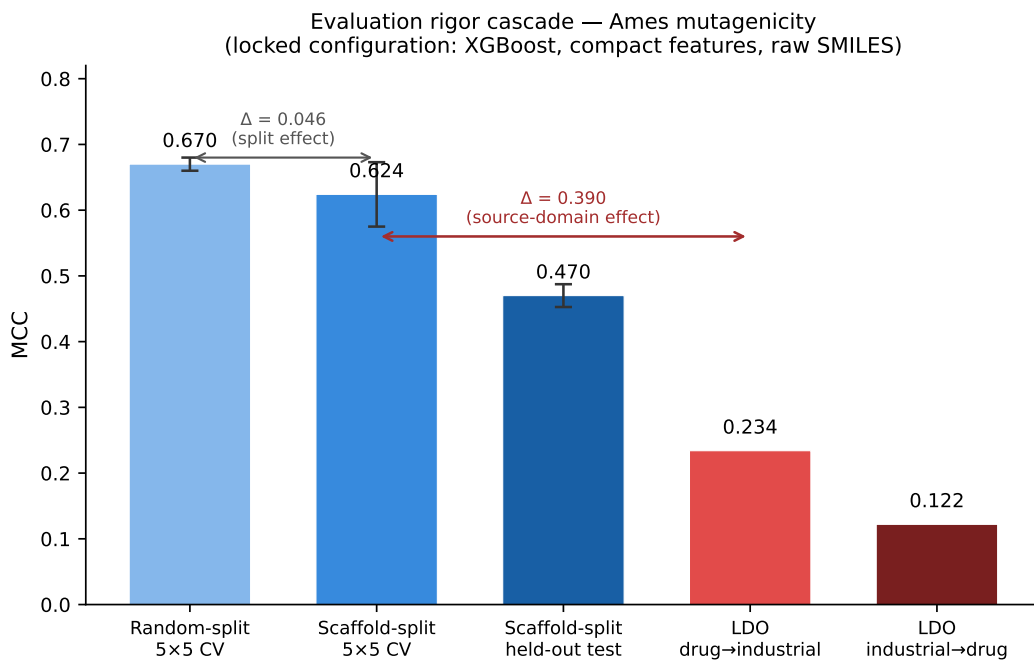


Figure 3: Evaluation rigor cascade for Ames mutagenicity (locked configuration: XGBoost, compact features, raw SMILES). MCC decreases from random-split CV (0.670) through scaffold-split CV (0.624) and scaffold-split held-out test (0.470) to cross-source-domain LDO (0.234 drug→industrial, 0.122 industrial→drug). The source-domain effect ($\Delta = 0.390$, scaffold CV to LDO) substantially exceeds the split-method effect ($\Delta = 0.046$). All values from the same model, features, and preprocessing.

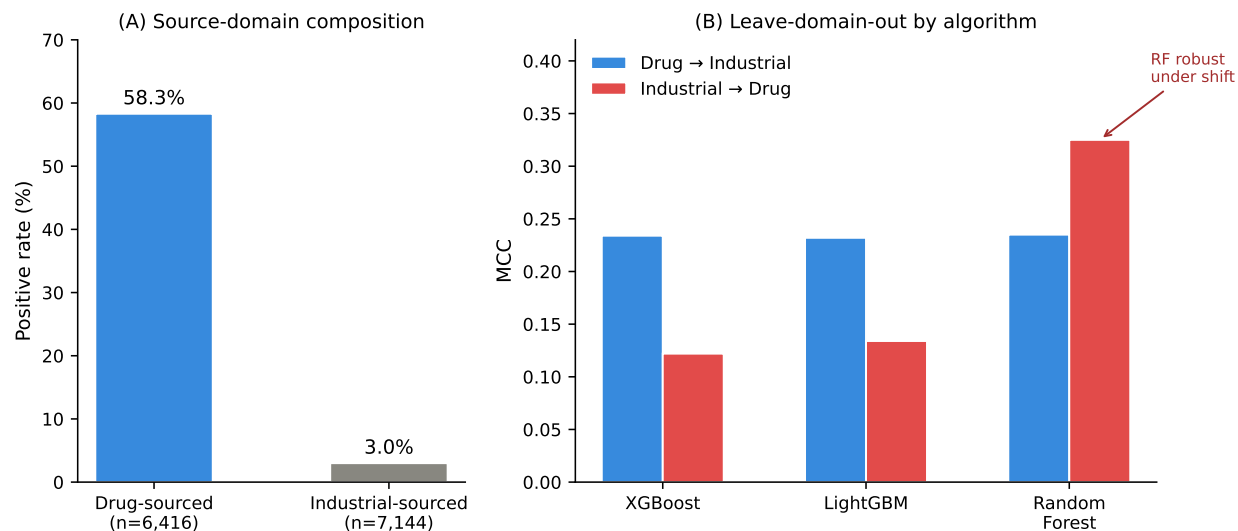


Figure 4: Source-domain composition and cross-domain robustness in the Ames dataset. (A) Positive rate by source domain: drug-sourced 58.3% vs. industrial-sourced 3.0% (19.5-fold difference). (B) Leave-domain-out MCC for three algorithms. Drug→industrial generalization is consistent across algorithms, but industrial→drug direction reveals Random Forest’s substantially better robustness (MCC = 0.325 vs. XGBoost 0.122).

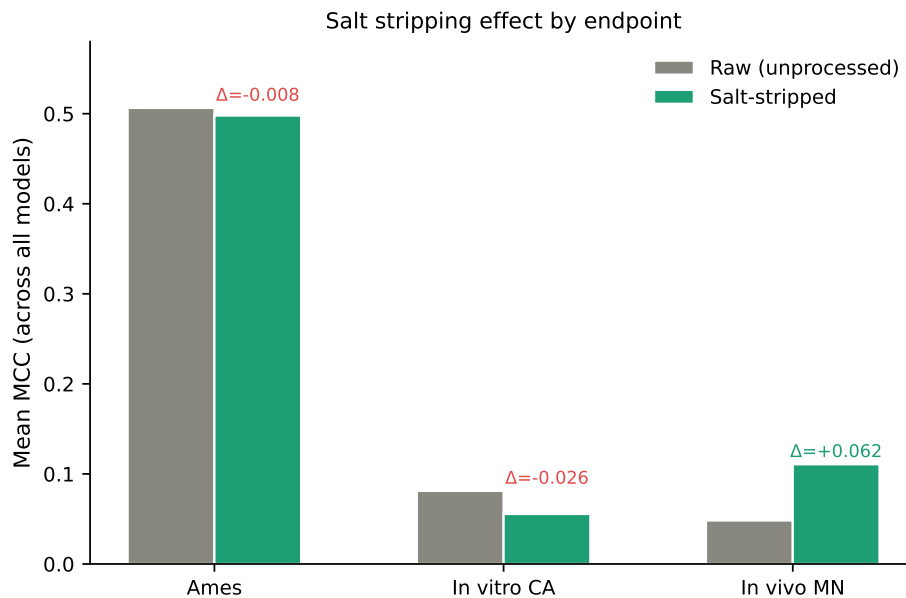


Figure 5: Effect of salt stripping on mean MCC (averaged across all models and feature modes) by endpoint. Salt stripping improved in vivo mean MCC from 0.048 to 0.111 ($\Delta = +0.063$); the best individual in vivo model showed a larger improvement of +0.225 (raw MCC = 0.086 → salt-stripped MCC = 0.311). Ames showed mixed effects; in vitro showed a decrease.

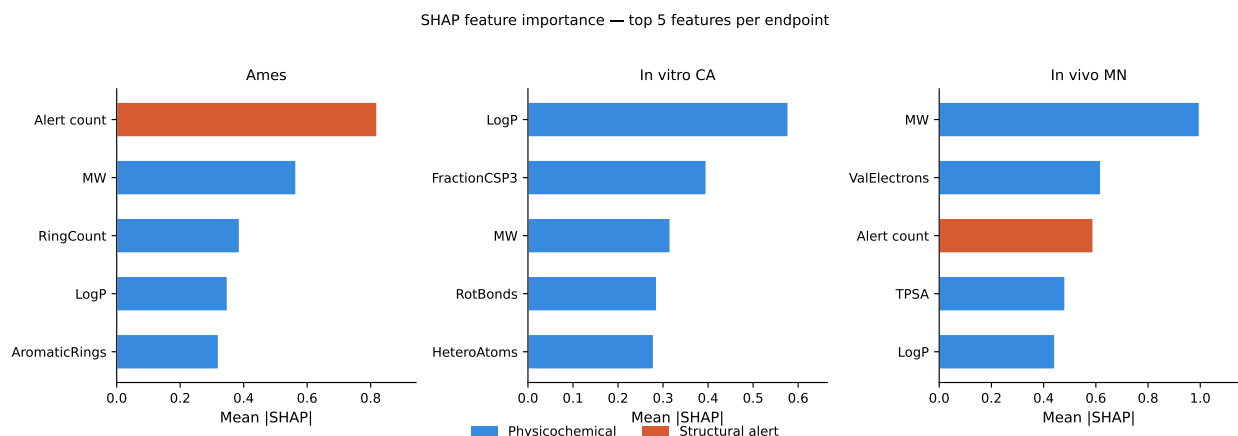


Figure 6: SHAP feature importance (top 5 features) for each primary endpoint, computed from the best-performing tree-based model. Physicochemical descriptors (blue) dominate across all endpoints. Structural alert count (orange) ranks first for Ames but is absent from the in vitro top 5. Individual fingerprint bits are absent from the top 5 for all endpoints despite comprising 50–75% of features in the broad modes.

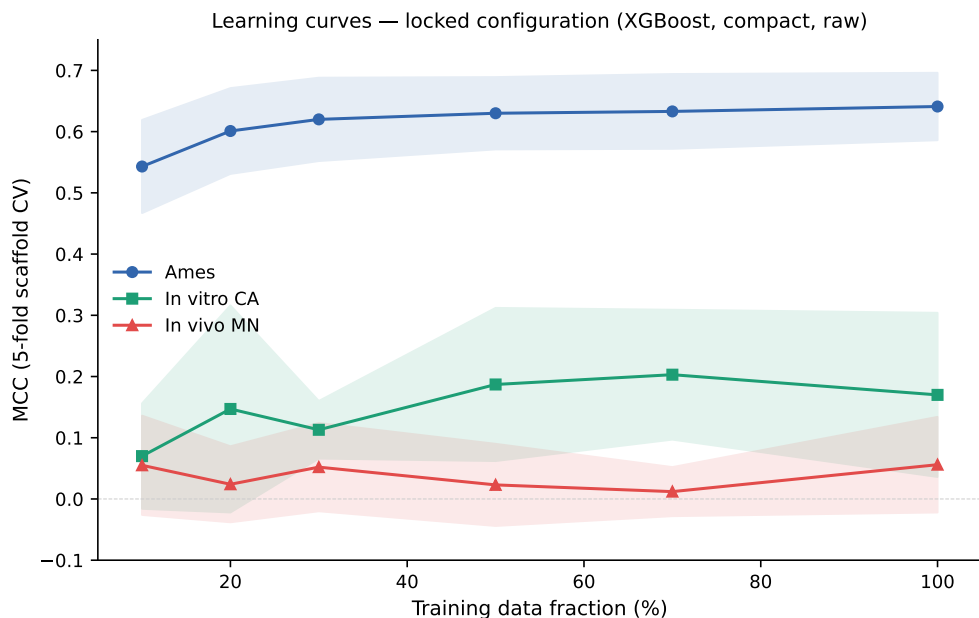


Figure 7: Learning curves for the locked configuration (XGBoost, compact features, raw SMILES). Ames MCC saturates by 30% of training data; in vitro shows a noisy upward trend; in vivo is flat throughout, indicating that additional data does not improve classification under this configuration. Shaded regions represent ± 1 SD across 5-fold scaffold CV. Note: the in vivo curve reflects the locked configuration only; the best-performing salt-stripped configuration was not evaluated.